Black Box Testing: Decision Table

Problem Statement: Consider Triangle problem : /* Design and develop a program in a language of your choice to solve the triangle problem defined as follows : Accept three integers which are supposed to be the three sides of triangle and determine if the three values represent an equilateral triangle, isosceles triangle, scalene triangle, or they do not form a triangle at all. Derive test cases for your program based on decision-table approach, execute the test cases and discuss the results */ and attached all printouts/screenshots/summary report here

Conditions:
- $a < b + c$?
- $b < a + c$?
- $c < a + b$?
- $a = b$?
- $a = c$?
- $b = c$?

(first three conditions: a, b, c, form a triangle?)

Actions:
- Not a Triangle
- Scalene
- Isosceles
- Equilateral
- Impossible

ii. Make initial decision table consisting of columns (using the formula 2^conditions)

iii. Reduced the table to remove redundant/impossible scenarios.

iv. write test cases for reduced decision table

v. execute code for triangle problem and execute test cases (iii)

Conditions:

1. $a < b + c$
2. $b < a + c$
3. $c < a + b$
4. $a = b$
5. $a = c$
6. $b = c$

Actions:

1. not a triangle
2. scalene triangle
3. isosceles triangle
4. equilateral triangle
5. no decision

columns : 2^6 = 64

Initial Table:

CONDI-TIONS	**R1**	**R 2**	**R 3**	**R 4**	**R 5**	**R 6**	**R 7**	**R 8**	**R 9**	**R 10**	**R 11**	**R 12**	**R 13**	**R 14**	**R 15**	**R 16**	**R 17**	**R 18**	**R19 - 64**

C1	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**DO NOT CARE** **(as the second pre condition of b < a + c is false therefore the action will be same)**
C2	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**	**F**	
C3	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**T**	**T**	
C4	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**T**	**T**	
C5	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	
C6	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	
ACTONS																			
A1									**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**	
A2								**O**											
A3				**O**		**O**	**O**												
A4	**O**																		
A5		**O**	**O**		**O**														

Reduced Table 1

CONDI-TIONS	**R1**	**R2**	**R3**	**R4**	**R5**	**R6**	**R7**	**R8**	**R9**	**R10**	**R11**	**R12**	**R13**	**R14**	**R15**	**R16**	**R17**	**R18**
C1	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**
C2	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**	**X**
C3	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**F**	**X**	**X**
C4	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**X**	**X**
C5	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**X**	**X**
C6	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**X**	**X**
ACTONS																		
A1									**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**	**O**
A2								**O**										

A3				**O**		**O**	**O**											
A4	**O**																	
A5		**O**	**O**		**O**													

Reduced Table 2

CONDITION	**R1**	**R2**	**R3**	**R4**	**R5**	**R6**	**R7**	**R8**	**R9 + R10 + R11**	**R12 - 18**
C1	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**
C2	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**x**
C3	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**T**	**F**	**x**
C4	**T**	**T**	**T**	**T**	**F**	**F**	**F**	**F**	**T**	**x**
C5	**T**	**T**	**F**	**F**	**T**	**T**	**F**	**F**	**x**	**x**
C6	**T**	**F**	**T**	**F**	**T**	**F**	**T**	**F**	**x**	**x**
ACTONS										
A1									**O**	**O**
A2								**O**		
A3				**O**		**O**	**O**			
A4	**O**									
A5		**O**	**O**		**O**					

TEST SUMMARY:

Project Name:	Triangle Problem Decision
Test ID:	Solve_Triangle_DT_TESE99
Test Title:	Use decision table approach for triangle problem
Test Priority:	Low
Module Name:	Identify_Triangle
Test Data:	Enter the 3 Integer Value (a, b and c)

Designed By:	Jassica
Designed Date:	2/16/2025
Executed By:	Jassica
Executed Date:	2/16/2025
Description of Test:	Check whether the given values form an equilateral, isosceles, scalene triangle or no triangle using equivalence
Pre-Conditions:	Range in which testing is to be done: $1 \le a \le 10$, $1 \le b \le 10$ and $1 \le c \le 10$ and $a < b + c$, $b < a + c$ and $c < a + b$
Dependencies:	No dependencies in this test.

Test cases:

TC-ID	TEST CASE DESCRIPTION	INPUT DATA			EXPECTED OUTPUT	ACTUAL OUTPUT	STATUS
		A	B	C			
R-01	All sides of inputs are equal	5	5	5	Equilateral	equilateral	pass
R-04	Two sides are same and one side is given different	5	5	1	isosceles	isosceles	pass
R-06	Two sides are same and one side is given different (i.e A and C are 5 and B is 1	5	1	5	Isosceles	isosceles	pass
R-07	Two sides are same and one side is given different (i.e B and C are 5 and A is 1	1	5	5	Isosceles	isosceles	pass

R-08	All three sides are different	10	9	5	scalene	scalene	pass
R-09	Two sides are same and one side is given different (i.e A and C are 5 and B is out of range	5	10	5	Out of range	Range cannot be formed	pass

Execution:

R-01

```
Enter Value For A B C : 5
5
5
Equilateral triangle
```

R-04

```
Enter Value For A B C : 5
5
1
Isosceles triangle
```

R-06

```
Enter Value For A B C : 5
1
5
Isosceles triangle
```

R-07

```
Enter value for A B C: 1
5
5
Isosceles triangle
```

R-08

```
Enter Value For A B C : 5
9
5
Isosceles triangle
```

R-09

```
Enter value for A B C: 5
10
5
Out of range
```

2. Consider a library management system(LMS) which issues book(s) to a registered user by checking the outstanding fee i.e. if a registered user have no pending fee then requested book may be issued(if and only if the potential user has under borrow limit)

i. Make initial decision table consisting of columns (using the formula 2^conditions)

ii. Reduced the table to remove redundant/impossible scenarios.

iii. write test cases for reduced decision table

iv. execute code for LMS problem and execute test cases (iii)

Design and develop a program in a language of your choice to develop above mentioned problem and attached all printouts /screenshots/summary report here.

(i) step 1: Identifying conditions

3 conditions:

1. registered user
2. user has no outstanding fee
3. user is under borrow limit

step 2: Identifying possible actions

action 1: book is issued

action 2: book is not issued

step 3: Initial table

no of columns = 2^conditions

no of columns = 2^3 = 8 columns

Conditions	R1	R2	R3	R4	R5	R6	R7	R8
registered user	T	T	T	T	F	F	F	F
no fee	T	T	F	F	T	T	F	F
under limit	T	F	T	F	T	F	T	F
Actions								
Book is issued	✔							
Book is not issued		✔	✔	✔	✔	✔	✔	✔

(iii) Reduced Table

Conditions	R1	R2	R3+R4	R5+R6+R7+R8
registered user	T	T	T	F
no fee	T	T	F	X
under limit	T	F	X	X
Actions				
Book is issued	✔			
Book is not issued		✔	✔	✔

(iii) Test Cases

Test Case	Test Data	Expected Output	Actual Output	Status	Comment
TC-01	registered: yes no outstanding fee: yes under borrow limit: yes	Displays" book can be issued"	Displays "book can be issued"	pass	

TC-02	registered: yes no outstanding fee: yes under borrow limit: no	Displays" book cannot be issued"	Displays "book cannot be issued"	pass	
TC-03	registered: yes no outstanding fee: no under borrow limit: don't care	Displays" book cannot be issued"	Displays "book cannot be issued"	pass	
TC-04	registered: no no outstanding fee: don't care under borrow limit: don't care	Displays" book cannot be issued"	Displays "book cannot be issued"	pass	

(iv)

```cpp
#include <iostream>
#include <string>

using namespace std;

class LibraryManagementSystem {
private:
    bool registered;
    bool outstanding_fee;
    bool under_borrow_limit;

public:
    LibraryManagementSystem() {
        registered = false;
        outstanding_fee = false;
        under_borrow_limit = false;
    }

    // Function to get user input (Yes/No)
    bool getUserInput(string prompt) {
        string response;
        while (true) {
            cout << prompt << " (yes/no): ";
            cin >> response;
            if (response == "yes" || response == "YES") return true;
            if (response == "no" || response == "NO") return false;
            cout << "Invalid input. Please enter 'yes' or 'no'." << endl;
        }
    }

    // Function to determine if the book can be issued
    string issueBook() {
        if (!registered) {
            return "Not Issued: User is not registered!";
        } else if (outstanding_fee) {
            return "Not Issued: Outstanding fee exists!";
```


```cpp
    // Function to determine if the book can be issued
    string issueBook() {
        if (!registered) {
            return "Not Issued: User is not registered!";
        } else if (outstanding_fee) {
            return "Not Issued: Outstanding fee exists!";
```


```cpp
        } else if (!under_borrow_limit) {
            return "Not Issued: Borrow limit exceeded!";
        } else {
            return "Book Issued Successfully!";
        }
    }

    // Function to run the LMS system
    void run() {
        cout << "\n--- Library Management System ---\n" << endl;

        // Get user inputs
        registered = getUserInput("Are you a registered user?");
        if (registered) {
            outstanding_fee = getUserInput("Do you have any outstanding fees?");
            if (!outstanding_fee) {
                under_borrow_limit = getUserInput("Are you under the borrow limit?");
            }
        }

        // Print result
        cout << "\n" << issueBook() << "\n" << endl;
    }
};

// Main function
int main() {
    LibraryManagementSystem system;
    system.run();
    return 0;
```

Outputs:

TC-01

```
--- Library Management System ---

Are you a registered user? (yes/no): yes
Do you have any outstanding fees? (yes/no): no
Are you under the borrow limit? (yes/no): yes

Book Issued Successfully!


--------------------------------
Process exited after 7.638 seconds with return value 0
Press any key to continue . . .
```

TC-02

```
--- Library Management System ---

Are you a registered user? (yes/no): yes
Do you have any outstanding fees? (yes/no): no
Are you under the borrow limit? (yes/no): no

Not Issued: Borrow limit exceeded!


--------------------------------
Process exited after 31.49 seconds with return value 0
Press any key to continue . . .
```

TC-03

```
--- Library Management System ---

Are you a registered user? (yes/no): yes
Do you have any outstanding fees? (yes/no): yes

Not Issued: Outstanding fee exists!


--------------------------------
Process exited after 31.99 seconds with return value 0
Press any key to continue . . .
```

TC-04

```
--- Library Management System ---

Are you a registered user? (yes/no): no

Not Issued: User is not registered!


--------------------------------
Process exited after 2.897 seconds with return value 0
Press any key to continue . . .
```